Osptek Display

TFT LCD SPECIFICATION

Model No:

YDP458B002-V13

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1. Document Revision History :

DOCUMENT REVISION	DATE	DESCRIPTION	PREPARED BY
A0	2025-7-22	First Release.	

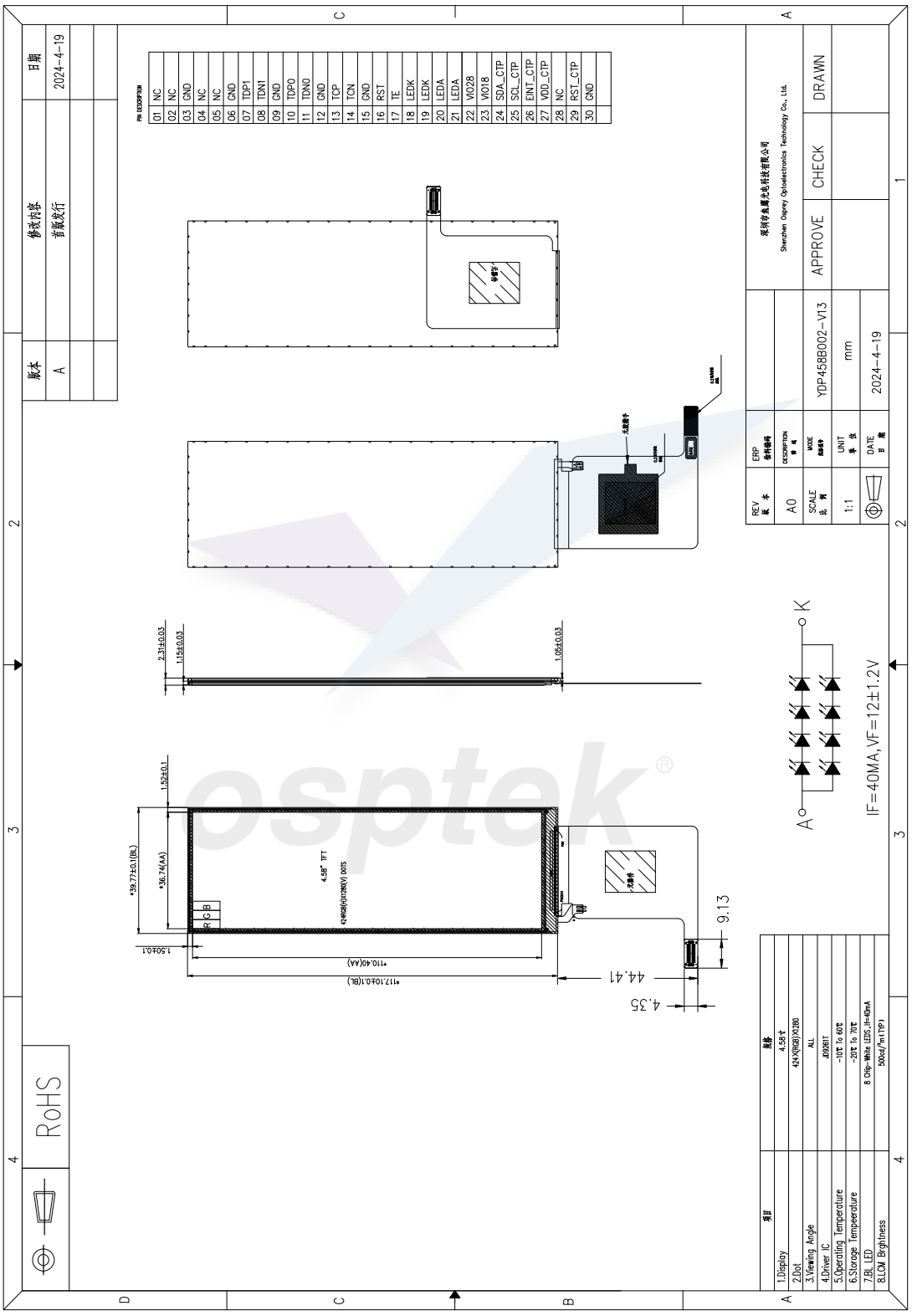
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2. General Description

No	Item	Specification	Remark
1	Screen Size	4.58inch	
2	Display Mode	Normally Black	
3	Resolution	424× RGB ×1280	
4	Active Area	36.74*110.4	mm
5	Outline Dimension	39.77*117.1*2.31	mm
6	Viewing Direction	ALL	
7	Driver IC	JD9261T	
8	Interface	MIPI	
9	Back Light	300	Cd/m^2

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3. Outline Dimension



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4. Interface Specification

Pin No	Symbol	Description	Note
1	NC	No Connection.	
2	NC	No Connection.	
3	GND	Ground.	
4	NC	No Connection.	
5	NC	No Connection.	
6	GND	Ground.	
7	D1P	Positive polarity of low voltage differential data 1 signal	
8	D1N	Negative polarity of low voltage differential data 1 signal	
9	GND	Ground.	
10	D0P	Positive polarity of low voltage differential data 0 signal	
11	D0N	Negative polarity of low voltage differential data 0 signal	
12	GND	Ground.	
13	CKP	Positive polarity of low voltage differential clock signal	
14	CKN	Negative polarity of low voltage differential clock signal	
15	GND	Ground.	
16	RST	Reset Signal input pin.（The levels of a and IOVCC are the same）	
17	TE	Frame head pulse for tearing effect.	
18-19	K	Power Supply For LED Backlight Cathode Input.	
20-21	A	Power Supply For LED Backlight Anode Input.	
22	VCC	Power Supply For LCD.	
23	IOVCC	Power Supply For I/O.	
24	CTP_SDA	Serial data input/output pin.	
25	CTP_SCL	Serial clock signal pin.	
26	CTP_INT	CTP interrupt request	
27	CTP_VCC	Power Supply For CTP	
28	NC	No Connection.	
29	CTP_RESET	Reset Signal input pin.	
30	GND	Ground.	

5. Absolute Maximum Ratings

Electrical Maximum Ratings – for IC Only

Parameter	Symbol	Min.	Max.	Unit	Note
Power supply voltage (VCI)	VCI	1.65	3.6	V	1
Power supply voltage (IOVCC)	IOVCC	1.65	3.6	V	1

Note:

IOVCC,VCI, GND must be maintained.

The modules may be destroyed if they are used beyond the absolute maximum ratings.

6. Electrical Specifications

Parameter	Symbol	Conditions	Min.	Typ.	Max.	Unit
Supply voltage (analog)	VCI -GND		1.65	-	3.6	V
Supply voltage (logic)	IOVCC -GND		1.65	-	3.6	V
Supply current (Logic & LCD)	ICC	-	-	-	-	mA
Supply voltage of white LED backlight	VLED	Forward urrent =40mA Number of LED =8	10.8	12	13.2	V

7. Timing Characteristics

11.3. AC characteristics

11.3.1. Reset input timings

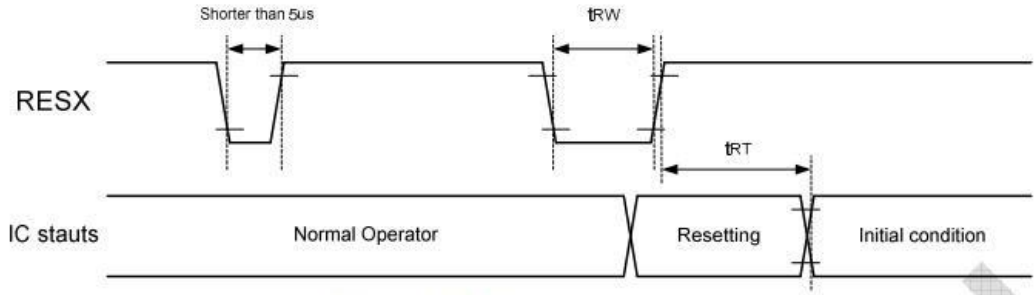


Figure 11.1: Reset input timings

Symbol	Parameter	Related pins	Min.	Max.	Unit
t_{RW}	Reset pulse width[2]	RESX	10	-	μs
t_{RT}	Reset complete time[3]	-	-	5 (Note 5)	ms
		-	-	120 (Note 6, 7)	ms

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11.3.2.TP Serial Interface Timing Characteristics (4-line SPI)

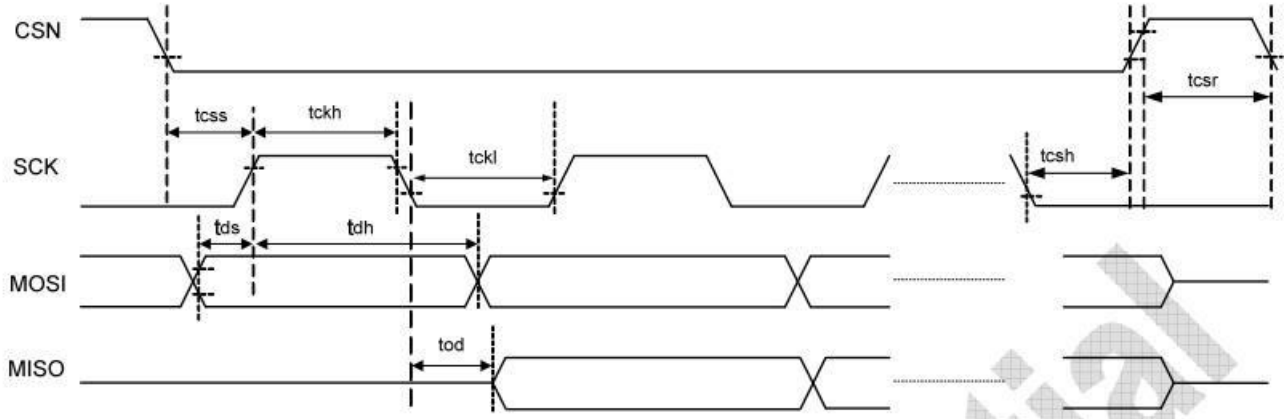


Figure 11.2: 4-line Serial Interface Timing Characteristics

Symbol	Parameter	Min.	Max.	Unit	Note
tcss	Chip select setup time	1000		ns	
tcsh	Chip select hold time	200		ns	
tcsr	Chip select recovery time	1000		ns	
tckh	Control pulse "H" duration	33.33		ns	Up to 15MHz
tckl	Control pulse "L" duration	33.33		ns	
tds	Data setup time	25		ns	
tdh	Data hold time	25		ns	
tod	Output delay time	-	50	ns	

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11.3.3.SPI Interface Timing Characteristics

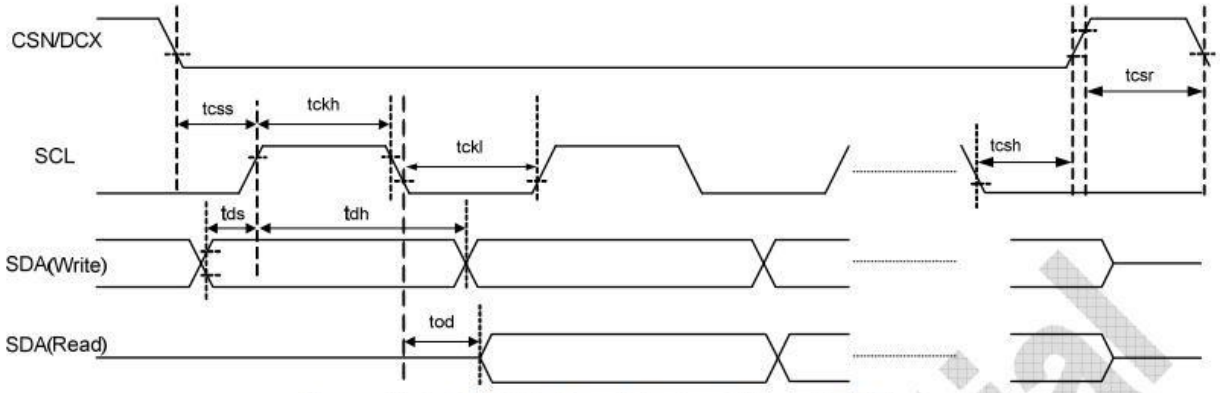


Figure 11.3: SPI Interface Timing Characteristics

Symbol	Parameter	Min.	Max.	Unit	Note
tcss	Chip select setup time	60		ns	
tcsh	Chip select hold time	15		ns	
tcsr	Chip select recovery time	40		ns	
tckh	Control pulse "H" duration	15		ns	
tckl	Control pulse "L" duration	15		ns	
tds	Data setup time	10		ns	
tdh	Data hold time	15		ns	
tod	Output delay time	30	-	ns	

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High-Speed Data-Clock Timing

This section specifies the required timings on the high-speed signaling interface independent of the electrical characteristics of the signal. The PHY is a source synchronous interface in the Forward direction. In either the Forward or Reverse signaling modes there shall be only one clock source. In the Reverse direction, Clock is sent in the Forward direction and one of four possible edges is used to launch the data.

The Master side of the Link shall send a differential clock signal to the Slave side to be used for data sampling. This signal shall be a DDR (half-rate) clock and shall have one transition per data bit time. All timing relationships required for correct data sampling are defined relative to the clock transitions. Therefore, implementations may use frequency spreading modulation on the clock to reduce EMI.

The DDR clock signal shall maintain a quadrature phase relationship to the data signal. Data shall be sampled on both the rising and falling edges of the Clock signal. The term “rising edge” means “rising edge of the differential signal, i.e. CLKP – CLKN, and similarly for “falling edge”. Therefore, the period of the Clock signal shall be the sum of two successive instantaneous data bit times. This relationship is shown in Figure 11.8.

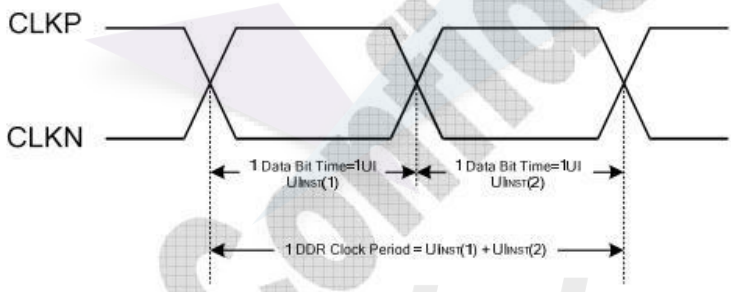
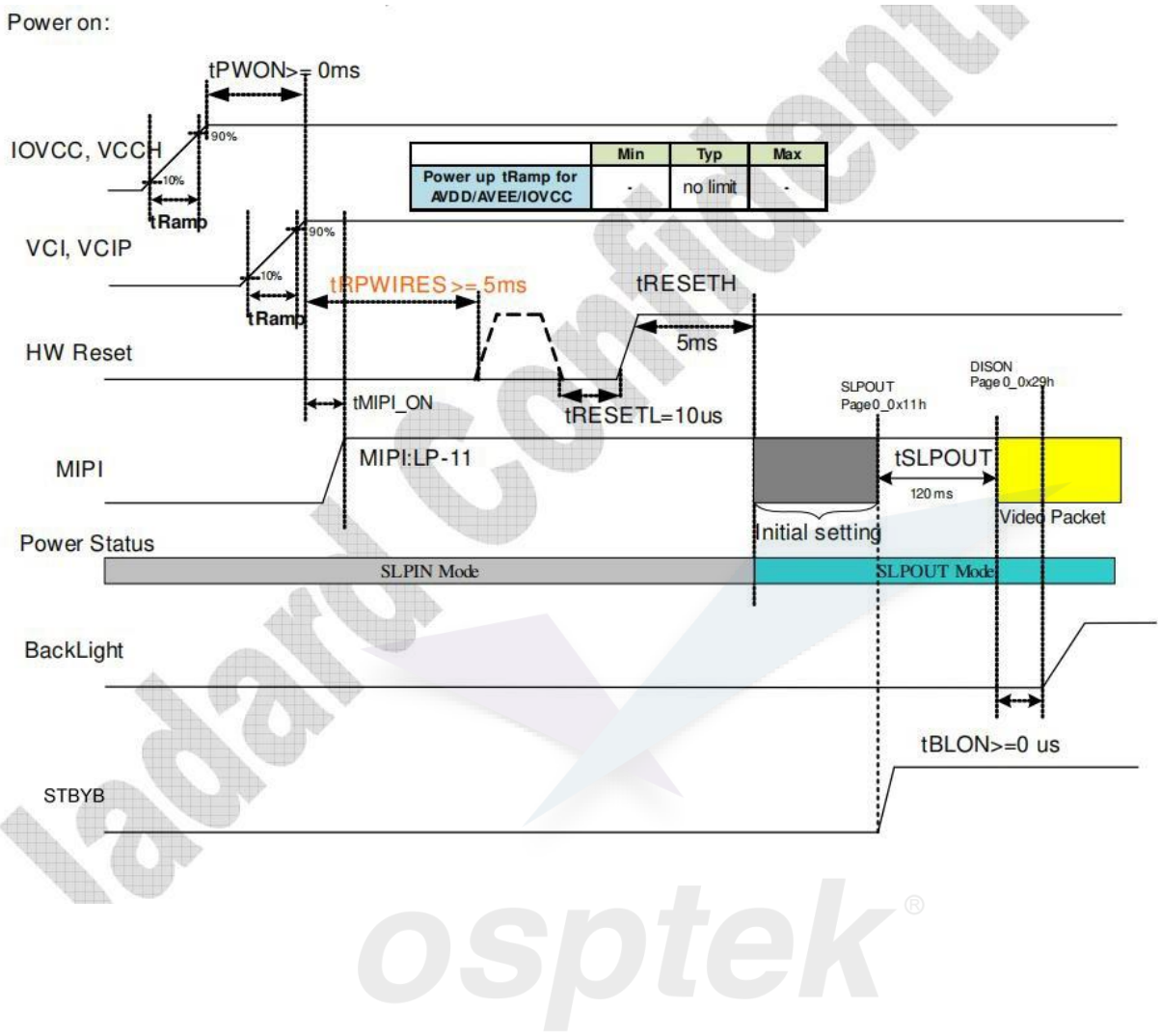


Figure 11.8: DDR Clock Definition

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8. Power Supply Configuration



	Min	Typ	Max
Power up tRamp for AVDD/AVEE/IOVCC	-	no limit	-

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9. Optical Specification

Item 项目	Symbol 符号	Condition 条件	Min 最小值	Typ 典型值	Max 最大值	Unit 单位	Note 备注
Response time 响应时间	Tr+Tf	Θ=0° Ø=0° Ta=25˚C	-	30	35	ms	1
Contrast ratio 对比度	Cr		800	1000	-	-	2
Color gamut 饱和度	S(%)		65	70	-	%	-
Luminance uniformity 均匀度	δWHITE		85	-	-	%	3
Viewing angle range 视角范围	Θx+	CR≧10 Ta=25˚C	75	80	-	deg	4
	Θx-		75	80	-	deg	
	Θy+		75	80	-	deg	
	Θy-		75	80	-	deg	
LCM Luminance LCM 亮度	Lv	Θ=0° Ø=0° Ta=25˚C	-	300	-	Cd/m²	5

Note1.Response time is the time required for the display to transition from White to black(Rise Time,Tr)and from black to white(Decay Time,Tf).For additional information see FIG1...

Note2.contrast Ratio(CR) is defined mathematically by the following formula ,For more information see FIG2.

Contrast Ratio(CR)=Average Surface Luminance with all white pixels/ Average Surface Luminance with all black pixels

Note3.The uniformity in surface luminance(WHITE) is determined by measuring luminance at eath test position,and then dividing the maximum luminance of all white pixels by minimum luminance of all white pixels,For more information seeFIG2.

WHITE=Minimum Surface Luminance with all white pixels(P1,P2,.....)/Maximum Surface Luminance with all white pixels(P1,P2, ...)

Note4.Viewing angle is the angel at which contrast ratio is greater than a specific value.For TET module,the specific value of contrast ratio is 10.For monochrome and color stn module,the specific value of contrast ratio is2.The angles are determined for the horizontal or x axis and the vertical or y axis with respect to the z axis which is normal to the LCD surface.For more information see FIG3

Note5. Surface luminance is the LCD surface luminance with all white pixels,For more information see FIG2.

LV=Average Surface Luminance with all white pixels(P1,P2,.....)

FIG1. The definition of Response time

响应时间定义

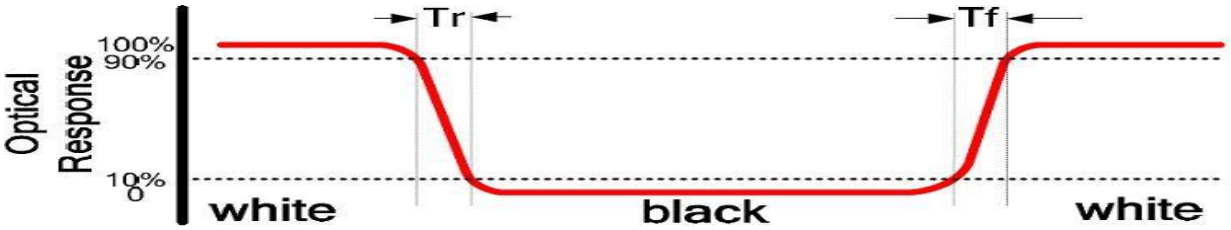


FIG2. Measuring method for Contrast ratio,surface luminance,Luminance uniformity,CIE(X,Y)chromaticity.

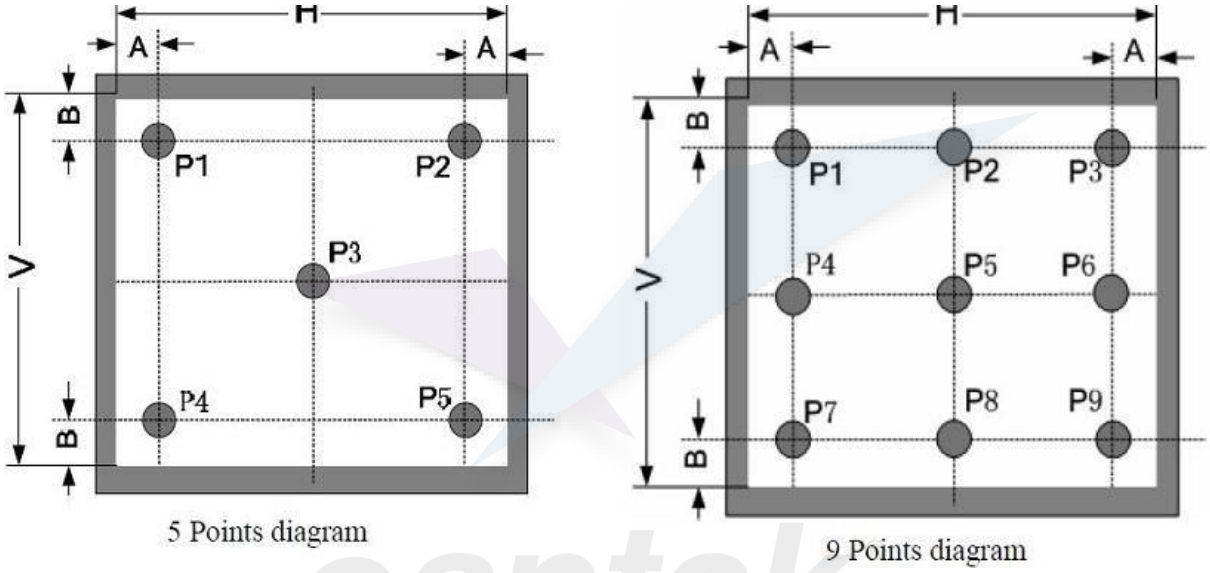
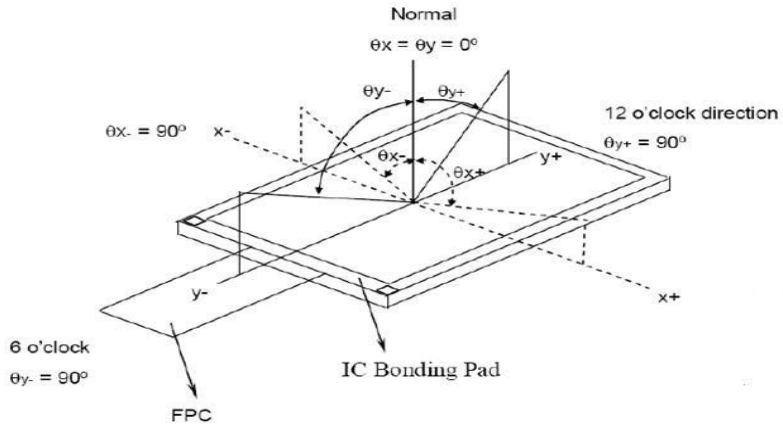


FIG3 The definition of viewing angle 视角定义



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10. Inspection Specifications

4.1 外观检查:

411 产品放≥600LUX 的日光灯环境下，眼睛距被检验物体 30±10cm;

412 检验视角为 LCD 平面法线方向±45°以内，在其他角度看见不算缺陷,客户特别要求除外。（见图二）

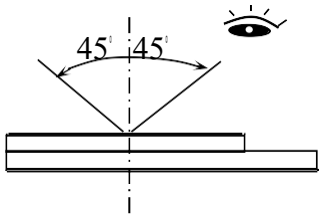
413 每片产品外观检验时间 5—10 秒;

4.2 电性检查

421 暗房亮度标准≤100LUX，眼睛距被检验物体 30±10cm;

422 按被检产品对应的《电性检查作业指导书》要求设定电测参数进行电性检验;

423 按《产品图纸》中的视角方向进行检验，为 LCD 平面法线方向±45°以内，客户特别要求除外。在其他角度看见不算缺陷（见图二）;



图二

4.3 抽样计划及允收水准:

431 采用正常检验一次抽样，GB/T2828.1-2012 一般检查水平Ⅱ，实验取样按 S-2 抽样计划数安排;

10.3.1.1 致命缺陷（CR） AQL 0

10.3.1.2 重缺陷（MA）: AQL 0.4

10.3.1.3 轻缺陷（MI）: AQL 0.65

432 加严检验:同一机种客户连续 3 次投诉或批退、或则客户上线不良超过 8000PPM、OQC 一月内同一机型连续三次批退；OQC 则按照 GB/T2828.1-2012 一般检查水平Ⅱ，正常抽样计划表，AQL CR=0 MA=0.25 MI=0.4。

433 放宽检验:三个月内同一机型连续交货 10 批以上客户端无重大投诉或则退货，OQC 则按照 GB/T2828.1-2012 一般检查水平Ⅱ，放宽抽样计划表，AQL CR=0 MA=0.65 MI=1.0 进行抽样。

4.4 计量单位

以下标准如无特别注明，单位均以 mm 计.

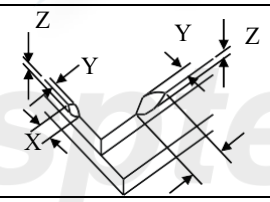
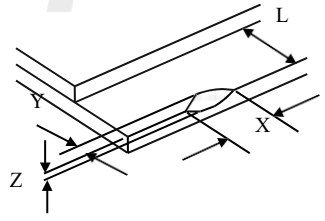
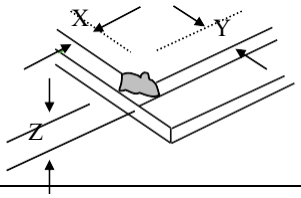
4.5 检验标准:

451 功能检验标准

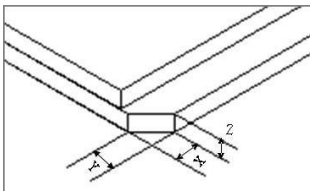
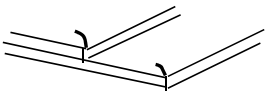
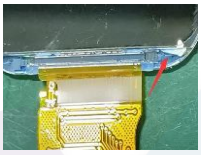
不良类型		检验方法及接收标准		判定	缺陷程度
点状	点状不良（电性点不良、外观点不良）	菲林卡	Φ≤0.1	不计，密集点不允许	MI
			0.1＜Φ≤0.15，间距 10MM	2	
			0.15＜Φ≤0.20	1	
			Φ＞0.20	0	
线状	显示功能线状不纳入	菲林卡	w＜0.02	不计	MA
			L≤2,0.02≤w≤0.03	2	
			L≤4,0.03≤w≤0.05	1	
			w>0.05	以点状规格判定	
显示状态	无显	目视	/	NG	MA
	黑屏		/	NG	
	乱码		/	NG	
	缺线		/	NG	
	暗缺线		/	NG	
	多缺线		/	NG	
	少画面		/	NG	
	屏闪		/	NG	
	显示浓/淡		/	NG，B、C 类依限度样板	
	发白		/	NG，B、C 类依限度样板	
	倒屏		/	NG	
	花屏		/	NG	
	显异		/	NG	
	漏液		/	NG	
	酒精印		/	NG	
	膜鼓		/	NG	
	灯亮不稳定/灯不亮		/	NG	
	显示暗		/	NG	

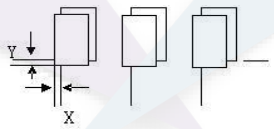
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	背光色差		/	放入B 规	
	单 LED 色差		/	放入B 规	
	反黑/白		/	NG	
	TP 无触摸/不灵敏		/	NG	
显示状态	TP 触摸跳点		/	NG	
	T 触摸键失灵/不灵敏		/	NG	
	显示不均		/	NG,B 类依限度样板/ND 8	
	少边框	目视	边框不完整	依限度样板	MA
	边缘漏光		边缘漏黄光 OK,漏白光以客户组装实机效果为准	NG	
	灯眼		距离产品 30CM,±45° 视角以内，看不到 OK,反之 NG	限度样板	
	亮边		轻微接受	B 类依限度样板	
	视角反		/	NG	
	膜鼓		/	NG	MA
	背光框胶变形		1. 进入显示区; 2. 边缘透光;	NG	
外观	粘附力不够	用手撕易撕贴撕不起保护膜		NG	
	易撕贴位置不正确	目视	1. 不符合图纸要求NG; 2. （有上铁框类产品）易撕贴有胶部分贴在上铁框边缘≤1MM，超出 NG	/	MI
	易撕贴歪斜		易撕贴与玻璃边缘角度偏离垂直方向大于 10 度	NG	
	保护膜表面脏污		笔迹、胶状物质、手指印	NG	MI
			油污、脏印	≤1/6	
			油污、脏印	≤1/4	

<table>
<tr><td></td><td>保护膜翘</td><td>目视</td><td>1.未进行显示区OK,
2.进入显示区引起偏光片表面脏污NG;</td><td>/</td><td></td></tr>
<tr><td></td><td>保护膜气泡</td><td>目视</td><td>Φ≤3mm</td><td>允许 4 个</td><td></td></tr>
<tr><td rowspan="8">外观</td><td>偏光片凹凸点
压痕</td><td>目视</td><td>1. 凹凸点依点状标准；
2. 压痕依线状标准；
3. 特殊情况依限度样放行；</td><td>/</td><td></td></tr>
<tr><td>偏光片表面脏污</td><td rowspan="8">目视/
菲林卡</td><td>不允许</td><td>NG</td><td></td></tr>
<tr><td rowspan="2">偏光片边缘汽泡</td><td>边框区</td><td>Φ≤黑边框宽度 1/2</td><td></td></tr>
<tr><td>可视区</td><td>参照点、线缺陷标准</td><td></td></tr>
<tr><td rowspan="2">偏光片贴偏位</td><td>1.超出玻璃外缘边框使片翘起
2.往内偏移＞1/2;</td><td>NG</td><td rowspan="4"></td></tr>
<tr><td>1.上铁框盖住偏光片</td><td>NG</td></tr>
<tr><td>偏光片贴错</td><td>不符合图纸与样品要求，如（细砂片贴成光片）</td><td>NG</td></tr>
<tr><td rowspan="2">LCD 崩边</td><td>  </td><td>X 不计
Y≤黑边框宽度
Z≤T</td></tr>
<tr><td>  </td><td><table><tr><td>X*Y</td><td>Z</td></tr><tr><td>≤4mm²</td><td>≤T</td></tr></table>不允许伤及ITO 引线</td></tr>
<tr><td>外观</td><td>LCD 崩角</td><td>  </td><td>X≤黑边框宽度
Y≤黑边框宽度
Z≤T</td><td></td></tr>
</table>

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				X*Y ≤4mm² / Z ≤T	
	LCD 裂痕	目视/菲林卡/卡尺		NG	
	胶框缺口		长 度 ：≤2MM 宽度：≤0.5MM 边缘无披锋、不影整体尺寸	OK	MI
	硅胶未盖住ITO 引线	目视	不允许	NG	MA
	银浆拉丝成线			NG	
	漏打银浆		不允许（两点任一个点漏打均不允许）	NG	
	银浆未搭上玻璃CF 面		不允许（两点任一个点未搭上玻璃CF 面均不允许）	NG	MI
				NG	
	银浆上偏光片		≤0.2MM（在保护膜表面不影响片表面	OK	
外观	银浆偏移LCD 设定点	目视	不允许（两点任一个点出现均不允许）	NG	MA
	硅胶偏高		高度超过偏光片表面及上片	NG	MI
	硅胶汽泡、针孔		Φ≤0.3	不计	MI
			0.3＜Φ≤0.5	3	
			Φ ＞0.5	NG	
	FPC 裂缝		1.FPC 裂缝伤及线路NG; 2.FPC 裂缝未伤及到线路及无放大现象OK	/	
	FPC 折痕		死折未伤及线路；	NG	

	FPC 刺伤/划伤		未伤及基材线路影响电性	OK	
	金手指折痕		折死/折断	NG	
	金手指断裂		/	NG	
	金手指氧化		氧化手指≤4 根，不影响上锡焊接；	OK	
	金手指表面脏污		油污、胶类脏污	NG	
	金手指上锡		插接式不允许； 焊接式允许有≤2 根；	/	
	加强板（连接器）		偏斜错位	NG	
			上锡	NG	
			损坏	NG	
			规格不符	NG	
外观	FPC 补强板颜色不符		不符合样品	NG	
	焊接偏位	卡尺		X≤1/2 金手指宽度 Y≤0.5	
	锡珠、锡渣	目视	可移动性	NG	
			≤0.2MM（必须能被黄胶纸覆盖）	≤2 个	
			连接焊盘金手指	NG	
			超出黄胶纸覆盖范围	NG	
	锡尖		任何焊点或产品其它部位有锡尖存在	NG	
	锡高	卡尺	焊锡厚度	≤0.4mm；若工程文件有特殊要求则以特殊要求管控	
	焊接洁净度	目视	焊盘周围 3MM 范围内有明显助焊剂残留,触摸有粘性	NG	
			焊盘、金手指间助焊剂明显发黑	NG	

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			焊盘周围 1MM 范围内少量透明助焊剂残留	不计	
	元器件漏件、多件、错件 立件、虚焊		不允许	NG	MA
外观	元器件贴偏移位	目视	元件上下偏位≤元件长度的1/3 元件左右偏位≤元件宽度的1/3	/	
	背光混料	核对图纸与样品	与图纸/样品不符	NG	
	背光与FOG 分离	目视	用手撕易撕贴出现背光与LCD 分离	NG	
	胶框断裂		不允许	NG	
	背光 FPC 引线破裂		不允许	NG	MA
	胶框定位柱变形、破裂	目视	不允许	NG	
	铁框脱落		不允许	NG	
	铁框松动		松动不明显可接受	依限度样	
	铁框变形		超出规格尺寸	NG	
	生锈氧化		不允许	NG	
	铁框划伤		L 不计， W≤0.05	不计	
			L≤10，0.05＜W≤0.25	4	
			L≤3，0.25＜W≤0.35	2	
			L 不计，W＞0.35	0	
			喷码内容不符合图纸要求	NG	
			喷码内容歪斜、个别字体模糊不清	OK	

	TP 崩角/边		/	NG	MA
外观	TP 划伤	目视	依线状标准执行	/	MI
	TP 点不良/贴合点不良		依点状标准执行	/	
	TP 表面脏污		不明显的脏污可擦拭（需斜着灯光看）	OK	

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11. Reliability Test Items

Item	Test Condition	Criterion
High Temperature Storage	70 ℃, 96H	Note1,Note2
Low Temperature Storage	-20 ℃, 96H	
High Temp. & High Humidity Storage	60 ℃, 90％ RH, 96H	
Thermal Shock (Static)	-20℃, 30 min /70℃, 30 min, 20 cycles	
High Temperature Operation	60 ℃, 96H	
Low temperature Operation	-10 ℃, 96H	

Note1: Evaluation should be tested after storage at room temperature for two hours.

Note2:

Pass: Normal display image no line defect.

Fail: No display image, or line defects.

Partial transformation of the module parts should be ignored.

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12. Precautions

Please pay attentions to the followings as using the LCD module.

Handling

Do not apply strong mechanical stress like drop, shock or any force to LCD module. It may cause improper operation, even damage.

Because the polarizer is very fragile and easy to be damaged, do not hit, press or rub the display surface with hard materials.

Do not put heavy or hard material on the display surface, and do not stack LCD modules.

If the display surface is dirty, please wipe the surface softly with cotton swab or clean cloth.

Avoid using Ketone type materials (e.g. Acetone), Toluene, Ethyl acid or Methyl chloride to clean the display surface. It might damage the touch panel surface permanently. The recommended solvents are water and Isopropyl alcohol.

Wipe off water droplets or oil immediately.

Protect the LCD module from ESD. It will damage the LSI and the electronic circuit.

Do not touch the output pins directly with bare hands.

Do not disassemble the LCD module.

Do not lift the FPC of Touch Panel.

Storage

Do not leave the LCD modules in high temperature, especially in high humidity for a long time.

Do not expose the LCD modules to sunlight directly.

The liquid crystal is deteriorated by ultraviolet. Do not leave it in strong ultraviolet ray for a long time.

Avoid condensation of water. It may cause improper operation.

Please stack only up to the number stated on carton box for storage and transportation. Excessive weight will cause deformation and damage of carton box.

Operation

When mounting or dismounting the LCD modules, turn the power off.

Protect the LCD modules from electric shock.

The Driver IC control algorithms stated above should always obeyed to avoid damaging the LSI and electronic circuit.

Be careful to avoid mixing up the polarity of power supply for backlight.

Absolute maximum rating specified above has to be always kept in any case. Exceeding it may cause non-recoverable damage of electronic components or, nevertheless, burning.

When a static image is displayed for a long time, remnant image is likely to occur.

Be sure to avoid bending the FPC to an acute shape, it might break FPC.

Most of the touch screens have air vent to equalize the inside air pressure to the outside one. The air vent must be open and liquid contact must be avoided as the liquid may be absorbed if the liquid is accumulated near the air vent.

For the fragility of ITO film, it should avoid to use too tapering pen as the input material.

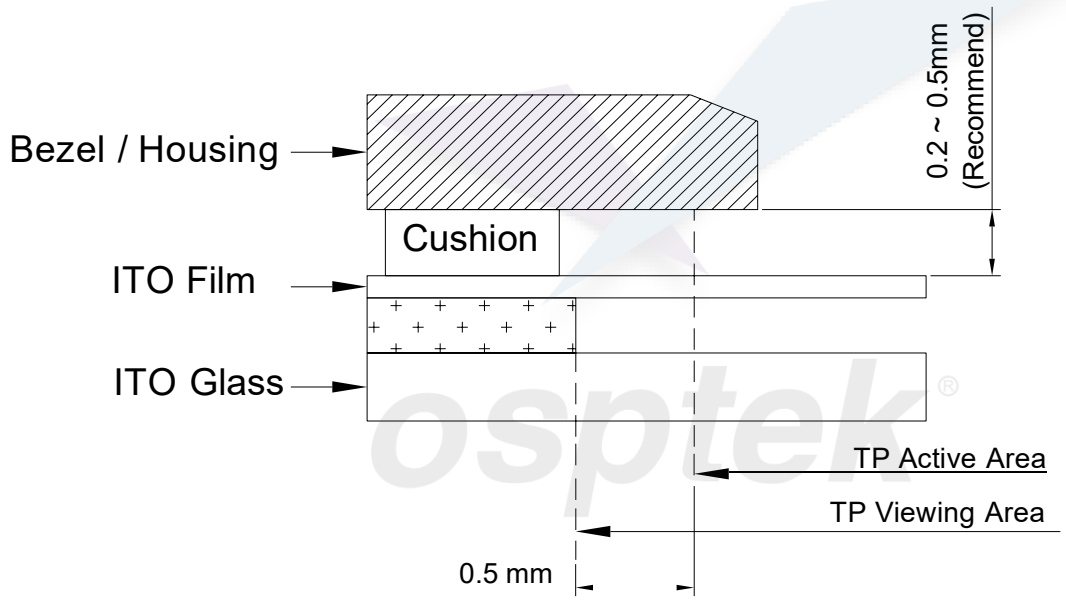
Touch Panel Mounting Notes

If a cushion is used between bezel/housing and film must be choose as free as enough to absorb the expansion and contraction to avoid the distortion of film.

The cushion must be placed out of the Viewing Area.

Bezel/Housing edge must be posited between Key Area and Viewing Area. The edge enters the Key Area may cause unexpected input if the gap is too narrow or foreign particles like dusts exist between Bezel/Housing and ITO film.

Mounting example:



The corner part has conductivity. Do not touch any metal part after mounting.

Others

If the liquid crystal leaks from the panel, it should be kept away from the eyes or mouth.

For the fragility of polarizer, it is recommended to attach a transparent protective plate over the display surface.

It is recommended to peel off the protection film on the polarizer slowly so that the electrostatic charge can be minimized.